

REQUIREMENT ANALYSIS PHASE

Overall Data Flow Diagram and User theory

Date	31 October 2025
Team ID	5C726FBC80DF9BA96244FB7712CD1C0D
Project Name	CRM Application for Jewel Management

1.Data Flow Diagram

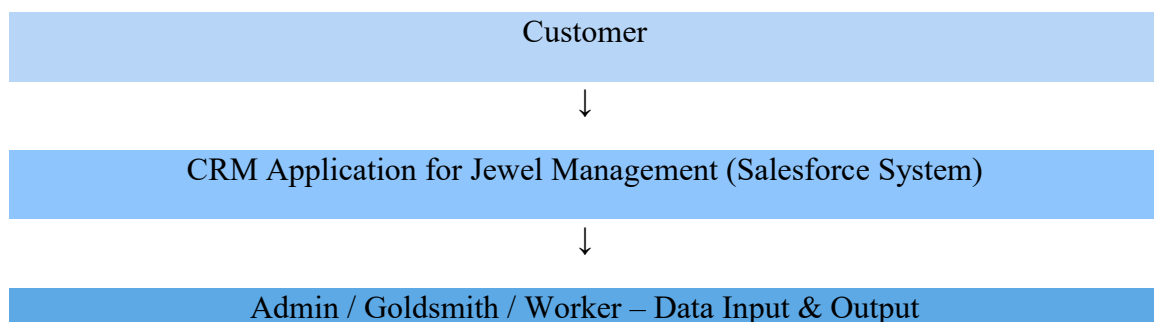
The Overall Data Flow Diagram (DFD) illustrates how data moves between users, processes, and the Salesforce CRM database in the CRM Application for Jewel Management. It demonstrates how customer, inventory, order, and billing data flow through various Salesforce modules, ensuring automation, real-time reporting, and secure data handling.

2. Level 0 – Context Diagram

The Level 0 Data Flow Diagram shows the main entities and their interactions with the CRM system. It represents the system as a single process interacting with external users.

- Admin monitors reports, dashboards, and workflows.
- Goldsmith updates item details, pricing, and billing.
- Worker manages stock and delivery records.
- Customer places orders and receives invoices.

Level 0 Data Flow Diagram:

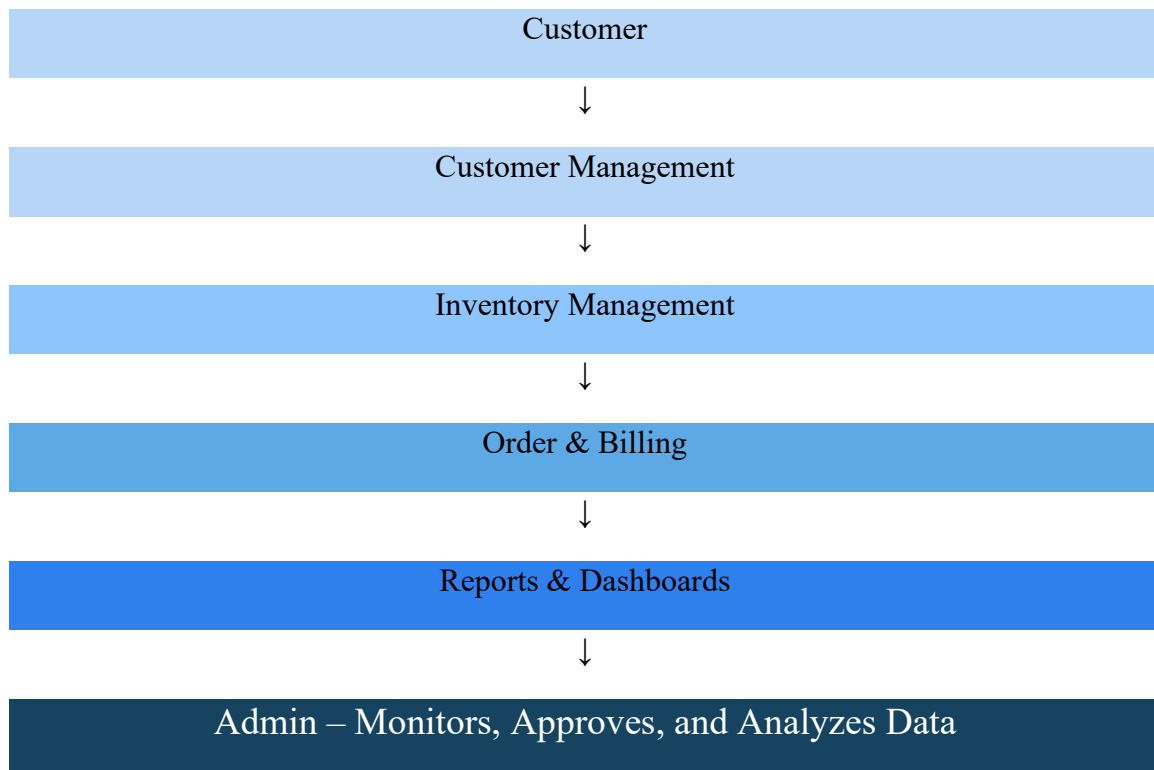


3. Level 1 – Detailed Data Flow Diagram

The Level 1 DFD expands the internal processes within the CRM system. It highlights how different Salesforce components interact to perform core business functions like customer management, inventory control, billing, and reporting.

- Customer Management – Stores and manages customer information.
- Inventory Management – Tracks stock levels and item details.
- Order & Billing – Processes orders, generates bills, and updates payment records.
- Reports & Dashboards – Generates analytics for the admin based on real-time data.

Overall Data Flow Diagram:



4. Data Storage and Automation

All modules interact with the Salesforce Database for real-time synchronization. Automation through Flows and Process Builder ensures that updates in one module (e.g., Billing) automatically reflect in other related modules. Reports and Dashboards utilize this unified data for analytical insights.

2. User Theories / User Role Explanation

The system includes multiple user roles with clearly defined permissions and responsibilities. Each role utilizes Salesforce CRM components in different ways.

Administrator

- Full system control with access to all modules.
- Creates and manages users, profiles, and roles.
- Monitors workflows, approvals, and automation.
- Generates comprehensive reports and dashboards for management.

Goldsmith

- Responsible for adding jewellery items, updating prices, and managing billing.
- Can create and modify Item, Price, and Billing records.
- Uses dashboards to view sales performance.
- Access controlled by the Goldsmith profile.

Worker

- Manages day-to-day operations such as updating stock quantities and handling customer orders.
- Has limited permissions to access Item and Customer Order objects.
- Cannot modify sensitive billing or pricing data.

Customer

- Interacts indirectly with the CRM through records maintained by staff.
- Their data (purchase, billing, and contact info) is stored in the Jewel Customer object.
- Used by Admin and Goldsmith for service and follow-up activities.

5. Conclusion

This multi-user design enables role-based data security and smooth collaboration. The Salesforce CRM's Role Hierarchy, Profiles, and Permission Sets ensure that each user can access only the features necessary for their role, maintaining both efficiency and safety in operations.